

Code for practical 11

```
import scala.collection.LinearSeq
import scala.io.Source
```

```
object Shreyash_S091_pract_11 {
  def main(args: Array[String]): Unit = {
    val filename = "Shreyash.txt"

    val lines = Source.fromFile(filename).getLines().toList

    val words = lines
      .flatMap(_.toLowerCase.split("\\W+"))
      .filter(_.nonEmpty)

    val wordCount = words.groupBy(identity).view.mapValues(_.size).toMap

    println("Word Frequencies:")
    wordCount.toSeq.sortBy(_._2).foreach { case (word, count) =>
      println(f"$word%-15s -> $count")
    }
  }
}
```

Output for practical 11

```
Word Frequencies:
am          -> 2
i           -> 2
science     -> 2
my          -> 2
in          -> 2
course      -> 1
bachelor    -> 1
manoj       -> 1
computer    -> 1
kadam       -> 1
second      -> 1
persueing   -> 1
hi          -> 1
dream       -> 1
shreyash    -> 1
currently   -> 1
year        -> 1

Process finished with exit code 0
```

Code for practical 12

```
import com.github.tototoshi.csv._
import java.io.File
```

```
object Shreyash_S091_P12 {
  def main(args: Array[String]): Unit = {

    val reader = CSVReader.open(new File("stock_data.csv"))
    val data = reader.allWithHeaders()
    reader.close()
  }
}
```

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S091

```
val categories = data.map(_("sector")).distinct.sorted

val newData = data.map { row =>

    val sector = row("sector")

    val oneHot = categories.map { cat =>
        cat -> (if (cat == sector) "1" else "0")
    }.toMap

    (row - "sector") ++ oneHot
}

val headers = newData.head.keys.toList
println(headers.mkString(", "))

newData.foreach { row =>
    println(headers.map(row).mkString(", "))
}

val writer = CSVWriter.open(new File("stock_encoded.csv"))

writer.writeRow(headers)

newData.foreach { row =>
    writer.writeRow(headers.map(row))
}

writer.close()

println("One-hot encoded file written to stock_encoded.csv")
}
```

Output for practical 12

```

open, low, date, Healthcare, close, Finance, day, volume, Technology, high
100.0, 99.09, 2026-01-01, 0, 100.89, 0, 1, 1914, 1, 101.44
100.89, 97.52, 2026-01-02, 0, 99.13, 1, 2, 3233, 0, 101.59
99.13, 96.54, 2026-01-03, 1, 97.09, 0, 3, 1895, 0, 100.47
97.09, 94.48, 2026-01-04, 0, 95.82, 0, 4, 3932, 1, 98.79
95.82, 94.99, 2026-01-05, 0, 96.77, 1, 5, 3413, 0, 98.36
96.77, 94.1, 2026-01-06, 1, 95.74, 0, 6, 1653, 0, 99.01
95.74, 95.01, 2026-01-07, 0, 96.94, 0, 7, 4920, 1, 98.12
96.94, 95.87, 2026-01-08, 0, 98.49, 1, 8, 2470, 0, 99.19
98.49, 96.78, 2026-01-09, 1, 100.48, 0, 9, 3988, 0, 102.19
100.48, 98.53, 2026-01-10, 0, 100.41, 0, 10, 1322, 1, 101.23
100.41, 98.99, 2026-01-11, 0, 100.84, 1, 11, 4529, 0, 103.0
100.84, 99.65, 2026-01-12, 1, 100.26, 0, 12, 3708, 0, 101.72
100.26, 98.35, 2026-01-13, 0, 98.96, 0, 13, 1953, 1, 101.34
98.96, 97.78, 2026-01-14, 0, 101.06, 1, 14, 4416, 0, 102.32
101.06, 99.68, 2026-01-15, 1, 100.49, 0, 15, 2093, 0, 102.3
100.49, 99.88, 2026-01-16, 0, 101.71, 0, 16, 3600, 1, 103.58
101.71, 99.37, 2026-01-17, 0, 100.12, 1, 17, 2554, 0, 103.67
100.12, 97.52, 2026-01-18, 1, 99.05, 0, 18, 1899, 0, 102.47
99.05, 97.38, 2026-01-19, 0, 100.18, 0, 19, 1938, 1, 102.36
100.18, 99.07, 2026-01-20, 0, 102.03, 1, 20, 1271, 0, 104.14
102.03, 98.84, 2026-01-21, 1, 100.65, 0, 21, 2288, 0, 104.42
100.65, 97.45, 2026-01-22, 0, 99.28, 0, 22, 3633, 1, 102.15
99.28, 98.74, 2026-01-23, 0, 99.21, 1, 23, 7288, 0, 100.71

```

Code for practical 13

```

import breeze.linalg._
import breeze.plot._
import com.github.tototoshi.csv._
import java.io.File

object Shreyash_S091_P13 {

  def main(args: Array[String]): Unit = {

    val reader = CSVReader.open(new File("stock_data.csv"))
    val data = reader.allWithHeaders()
  }
}

```

```

reader.close()

val technology = data.filter(_("sector") == "Technology")
val finance = data.filter(_("sector") == "Finance")
val healthcare = data.filter(_("sector") == "Healthcare")

def extractXY(rows: List[Map[String, String]]) = {
  val x = DenseVector(rows.map(_("open")).toDouble).toArray
  val y = DenseVector(rows.map(_("close")).toDouble).toArray
  (x, y)
}

val (xTechnology, yTechnology) = extractXY(technology)
val (xFinance, yFinance) = extractXY(finance)
val (xHealthcare, yHealthcare) = extractXY(healthcare)

val fig = Figure()
val plt = fig.subplot(0)

plt.title = "Open Price vs Close Price"
plt.xlabel = "Open Price"
plt.ylabel = "Close Price"

plt += plot(xTechnology, yTechnology, '.', name = "Technology", colorcode = "blue")

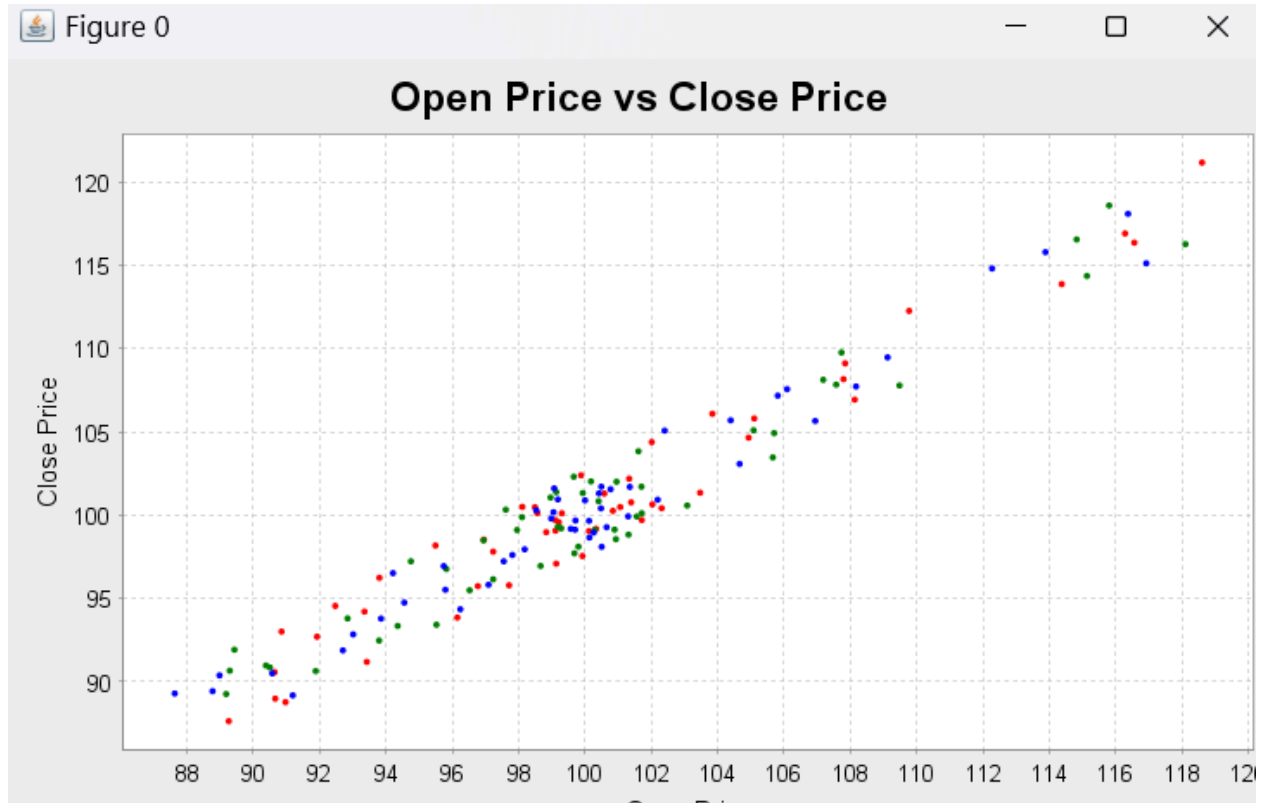
plt += plot(xFinance, yFinance, '.', name = "Finance", colorcode = "green")

plt += plot(xHealthcare, yHealthcare, '.', name = "Healthcare", colorcode = "red")

fig.refresh()
}
}

```

Output for practical 13



Code for practical 14

```
import breeze.linalg._
import breeze.plot._
import com.github.tototoshi.csv._
import java.io.File

object Shreyash_S091_P14 {

  def main(args: Array[String]): Unit = {

    val reader = CSVReader.open(new File("stock_data.csv"))
    val data = reader.allWithHeaders()

    reader.close()

    val closePrices =
      DenseVector(data.map(_("close").toDouble).toArray)
```

```
val fig = Figure("Histogram of Stock Closing Prices")

val binSizes = List(5, 10, 20)

for ((bins, idx) <- binSizes.zipWithIndex) {

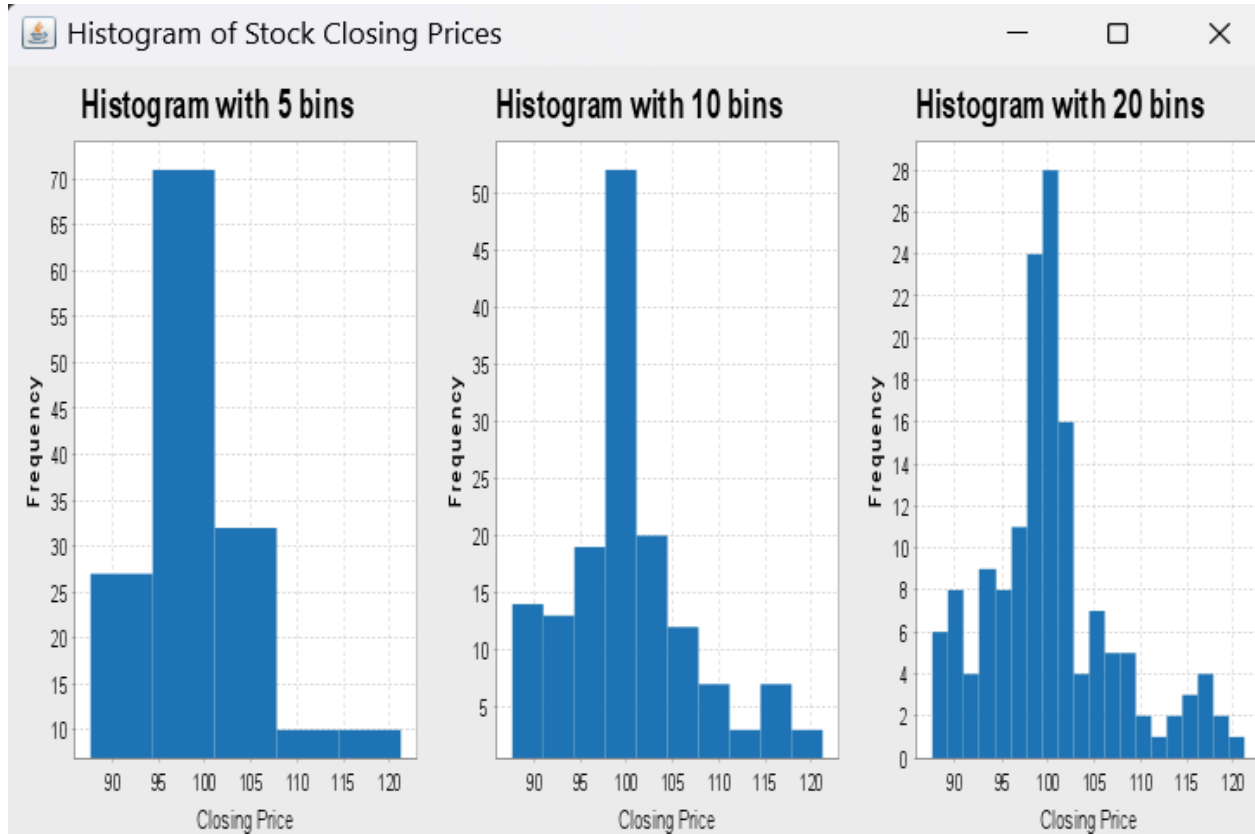
  val plt = fig.subplot(1, binSizes.length, idx)

  plt += hist(closePrices, bins)

  plt.title = s"Histogram with $bins bins"
  plt.xlabel = "Closing Price"
  plt.ylabel = "Frequency"
}

fig.refresh()
}
```

Output for practical 14



Code for practical 15

```
import breeze.linalg._
import breeze.plot._
import com.github.tototoshi.csv._
import java.io.File

object Shreyash_S091_P15 {

  def main(args: Array[String]): Unit = {

    val reader = CSVReader.open(new File("stock_data.csv"))
    val data = reader.allWithHeaders()

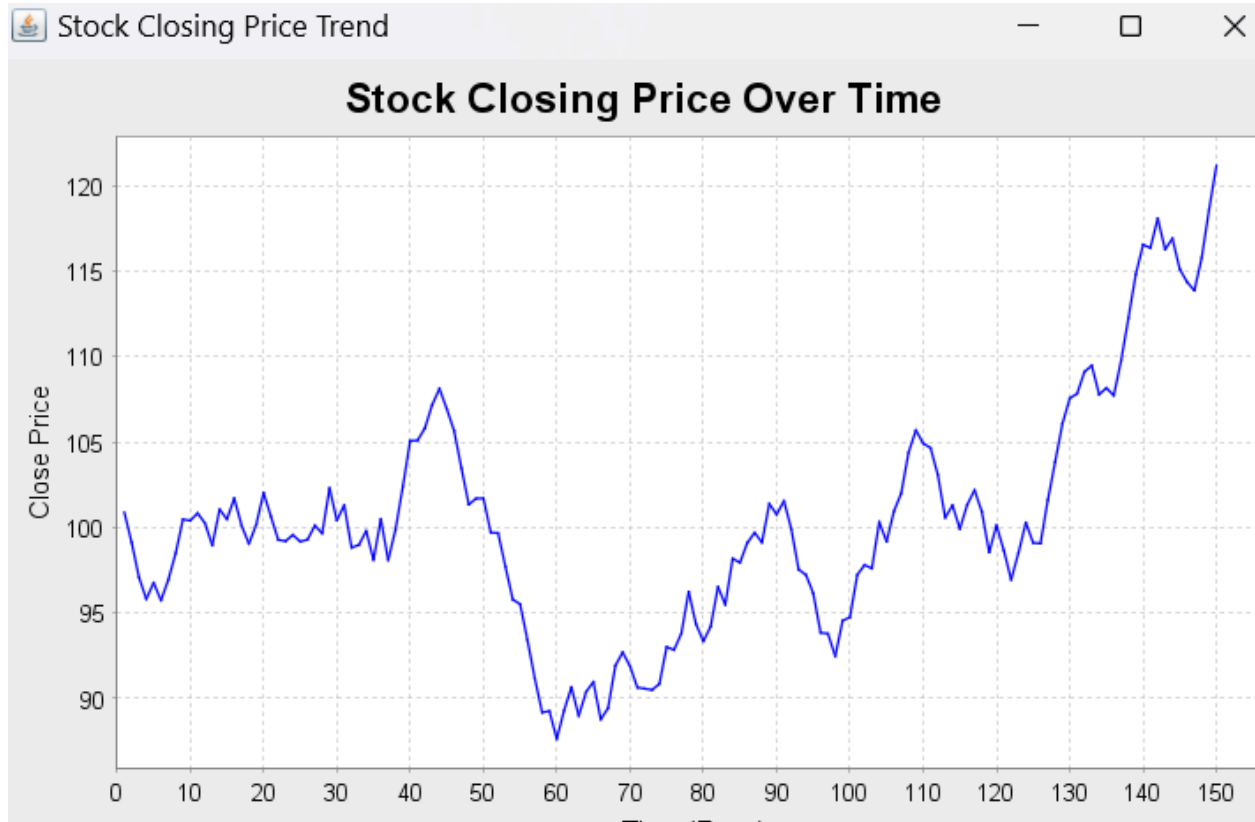
    reader.close()

    // Extract day and close price
    val x = DenseVector(
      data.map(_("day")).toDouble).toArray
```



```
)  
  
val y = DenseVector(  
  data.map(_("close").toDouble).toArray  
)  
  
val fig = Figure("Stock Closing Price Trend")  
val plt = fig.subplot(0)  
  
plt += plot(  
  x,  
  y,  
  name = "Close Price",  
  colorcode = "blue"  
)  
  
plt.xlabel = "Time (Days)"  
plt.ylabel = "Close Price"  
plt.title = "Stock Closing Price Over Time"  
  
fig.refresh()  
}  
}
```

Output for practical 15



Code for practical 16

```
import breeze.linalg._
import breeze.plot._
import com.github.tototoshi.csv._
import java.io.File

object Shreyash_S091_P16 {

  def main(args: Array[String]): Unit = {

    val reader = CSVReader.open(new File("stock_data.csv"))
    val data = reader.allWithHeaders()

    reader.close()
    val x = DenseVector(
      data.map(_("day")).toDouble).toArray
  )
}
```

```
val y = DenseVector(  
  data.map(_("close").toDouble).toArray  
)  
  
val fig = Figure("Stock Closing Price - Line + Scatter Plot")  
val plt = fig.subplot(0)  
  
plt += plot(x, y, name = "Close Price Line", colorcode = "blue")  
  
plt += plot(x, y, '.', name = "Close Price Points", colorcode = "red")  
  
plt.xlabel = "Time (Days)"  
plt.ylabel = "Close Price"  
plt.title = "Stock Closing Price - Line + Scatter Plot"  
  
fig.refresh()  
}  
}
```

Output for practical 16

